this page intentionally empty

1 Experimental Design and Results

1.1 Results

Table 1 shows the total execution time of the various encodings to execute the benchmark (`40 fibonacci`) for integer and floating-point values on the Apple M1.

Encoding	IntegerBr	Float
cachedPtr	3209ms	5083ms
compact	2977ms	2999ms
compact1	2913ms	3118ms
compact2	2977ms	3038ms
compact4	2977ms	3000ms
compact6	2978ms	3093ms
compactI	2852ms	2995ms
compactI1	2752ms	3044ms
compactI2	2847ms	3024ms
compactI4	2847ms	3050ms
compactI6	2872ms	3045ms
compactZ	2940ms	3069ms
nan	3033ms	3015ms
onlyFloat	N/A	2609ms
onlyInt	2659ms	N/A
ptr	4868ms	5194ms
spur	2742ms	3354ms
spurOpt	2724ms	3566ms
taggedHigh	3978ms	5359ms
taggedInt	2836ms	4829ms
taggedLow	N/A	N/A
taggedSMI	2757ms	4932ms
ultra	2760ms	3027ms
zagOrig	2969ms	3150ms
maxSpur	2744ms	3067ms

Table 1. AArch64 (Apple M1) total timing for encodings

Figures 1 and 2 show at a glance the relative performance of the various encodings on the two architectures.

All timings are the median of 10 runs following 3 runs of warmups. They are run on Linux systems running in console mode. We calculate the mean, median and geometric mean for each example and they are essentially equal in all cases.

Tables 2 and 3 present the net execution times for each encoding across the two benchmark configurations on AArch64 (Apple M1) and x86-64 respectively. Each row reports the time for a single encoding under two conditions: **IntegerBr** (inlined branching integer Fibonacci), and **Float** (inlined float Fibonacci).

1.1.1 Pareto Analysis.

The Pareto frontier contains the points that represent optimal trade-offs between two parameters. In this case, we are looking at integer versus floating-point performance in figure 5. The Pareto frontiers don't intersect, showing that the Aarch64 dominates the x86-64 in this experiment.

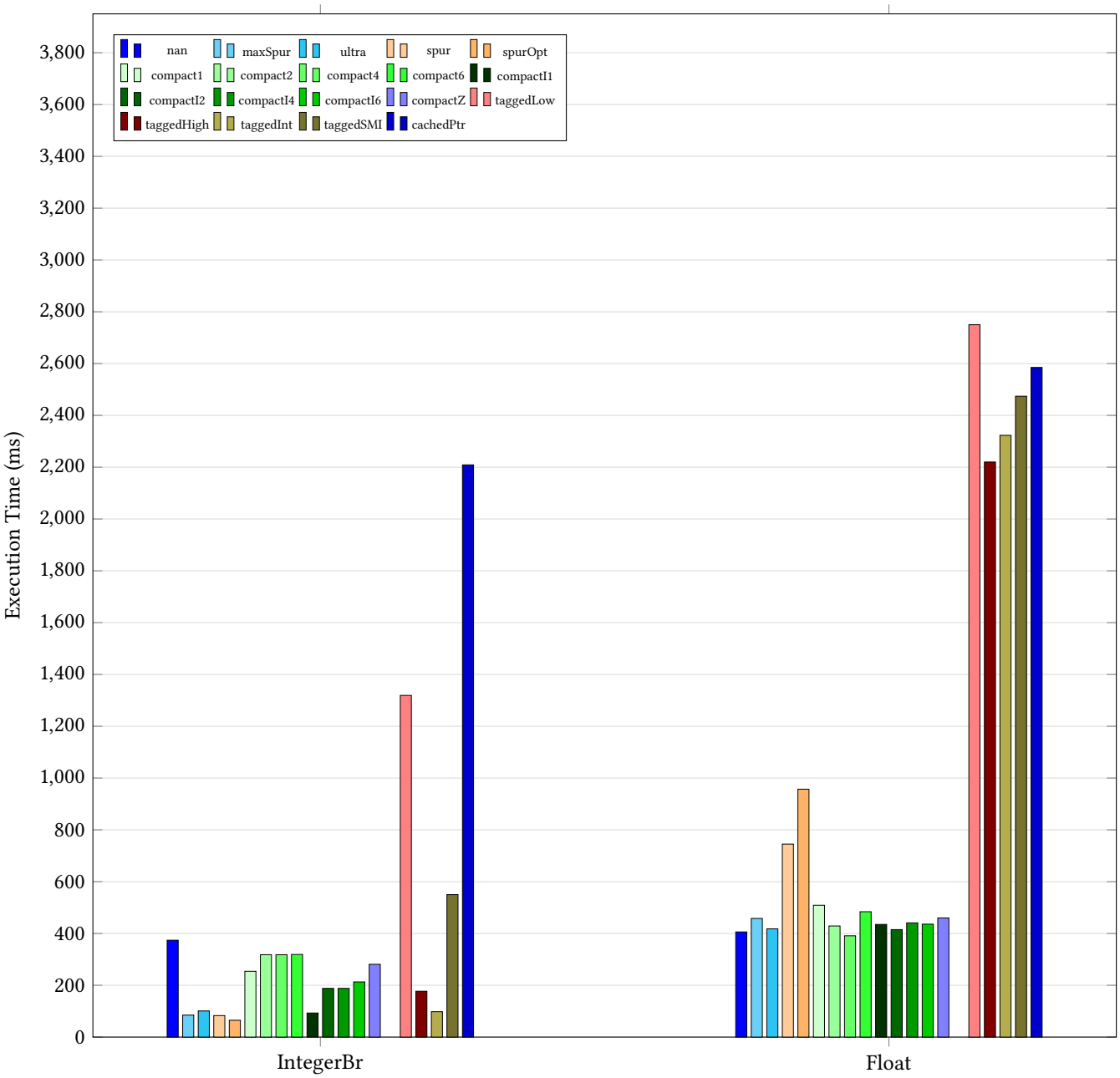

Figure 1. AArch64 benchmark results

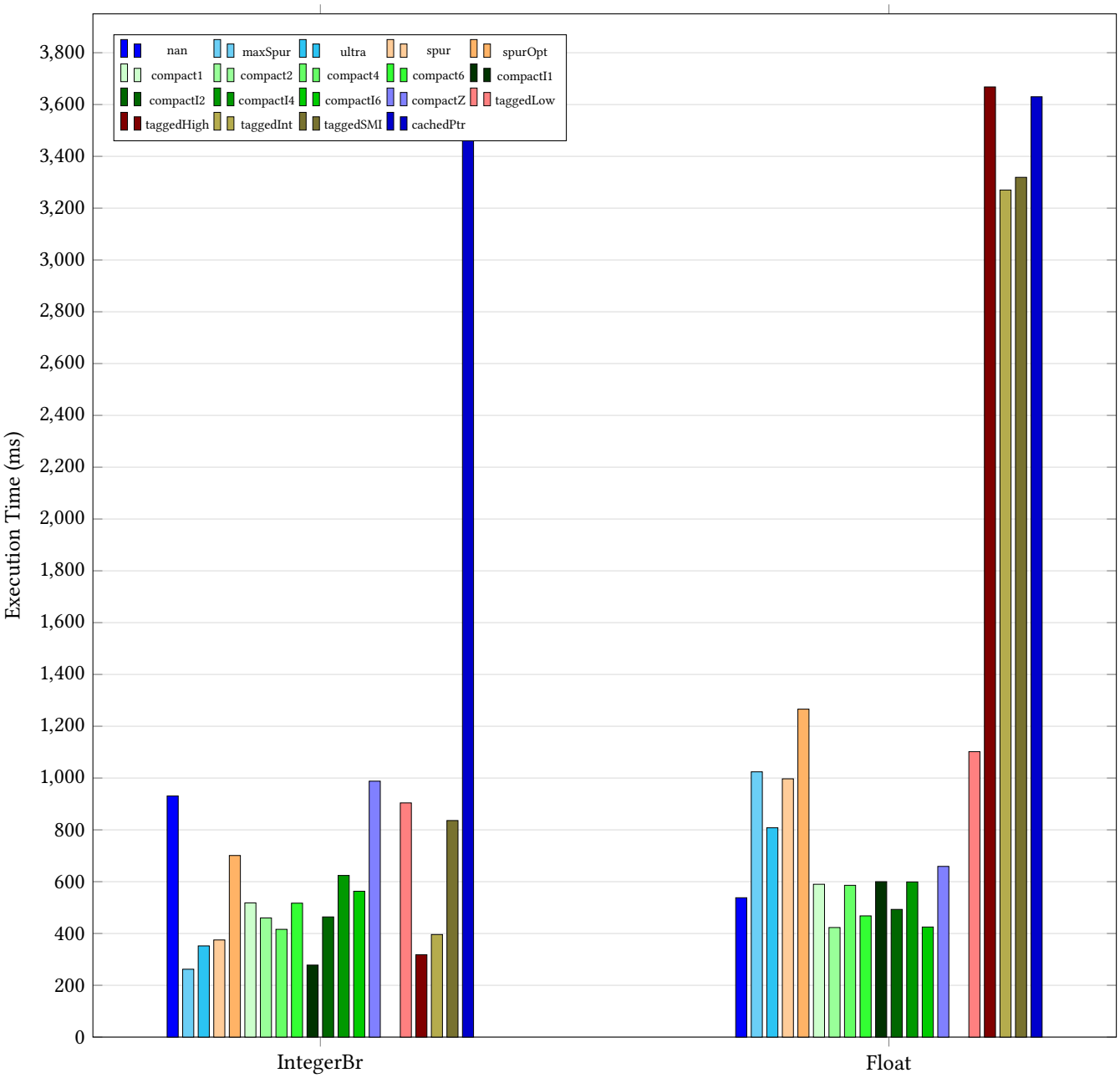

Figure 2. x86_64 benchmark results

Encoding	IntegerBr	Float
nan	374ms	406ms
maxSpur	85ms	458ms
ultra	101ms	418ms
spur	83ms	745ms
spurOpt	65ms	957ms
compact1	254ms	509ms
compact2	318ms	429ms
compact4	318ms	391ms
compact6	319ms	484ms
compactI1	93ms	435ms
compactI2	188ms	415ms
compactI4	188ms	441ms
compactI6	213ms	436ms
compactZ	281ms	460ms
taggedLow	N/A	N/A
taggedHigh	1319ms	2750ms
taggedInt	177ms	2220ms
taggedSMI	98ms	2323ms
cachedPtr	550ms	2474ms
ptr	2209ms	2585ms

Table 2. AArch64 (Apple M1) net timing for encodings

Encoding	IntegerBr	Float
nan	931ms	538ms
maxSpur	262ms	1024ms
ultra	352ms	808ms
spur	375ms	997ms
spurOpt	701ms	1266ms
compact1	518ms	590ms
compact2	460ms	423ms
compact4	416ms	586ms
compact6	517ms	468ms
compactI1	278ms	600ms
compactI2	464ms	493ms
compactI4	624ms	599ms
compactI6	563ms	425ms
compactZ	988ms	659ms
taggedLow	N/A	N/A
taggedHigh	904ms	1102ms
taggedInt	318ms	3668ms
taggedSMI	396ms	3270ms
cachedPtr	836ms	3319ms
ptr	3471ms	3630ms

Table 3. x86-64 (Linux) net timing for encodings

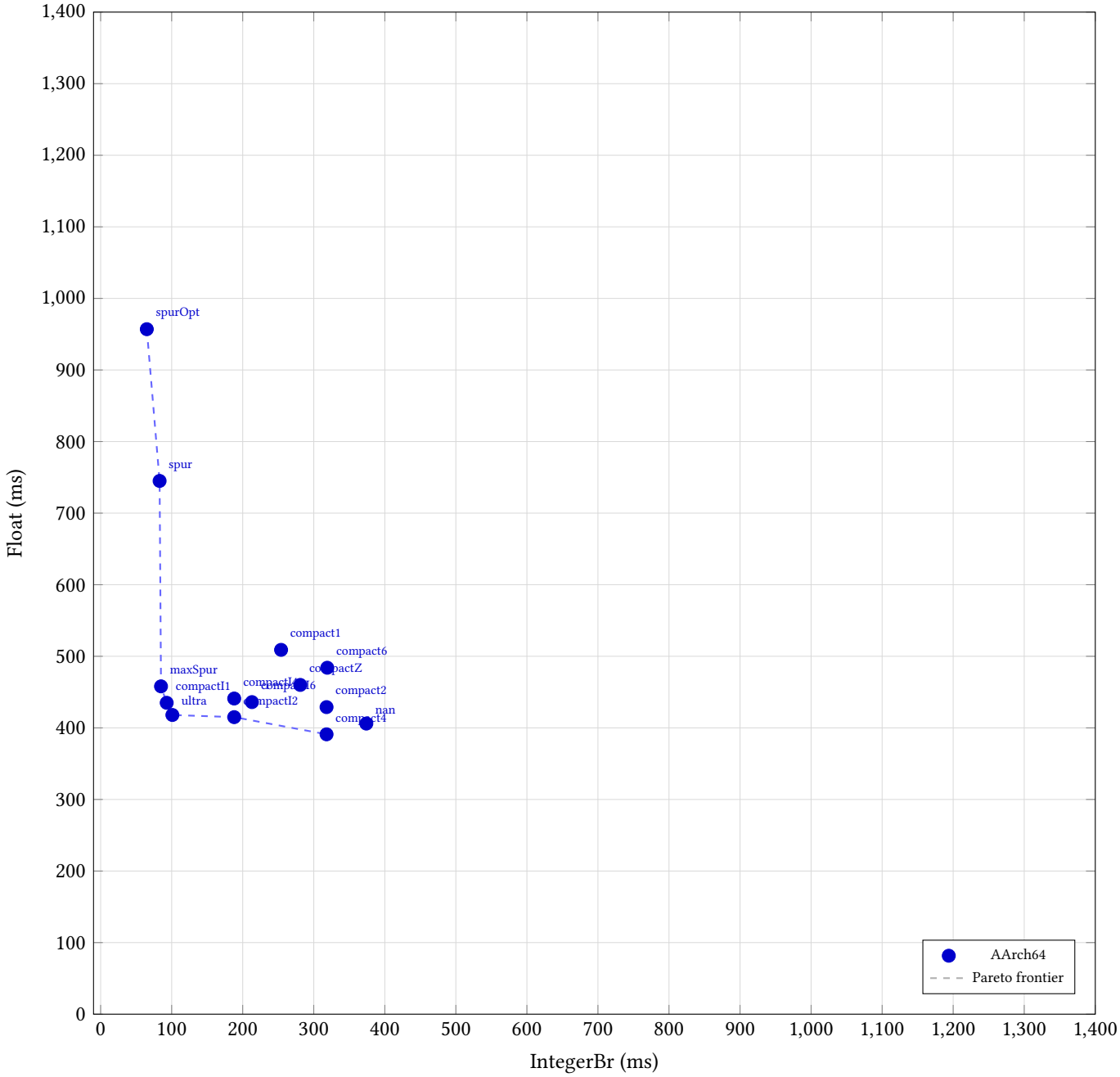

Figure 3. IntegerBr vs Float - AARCH64

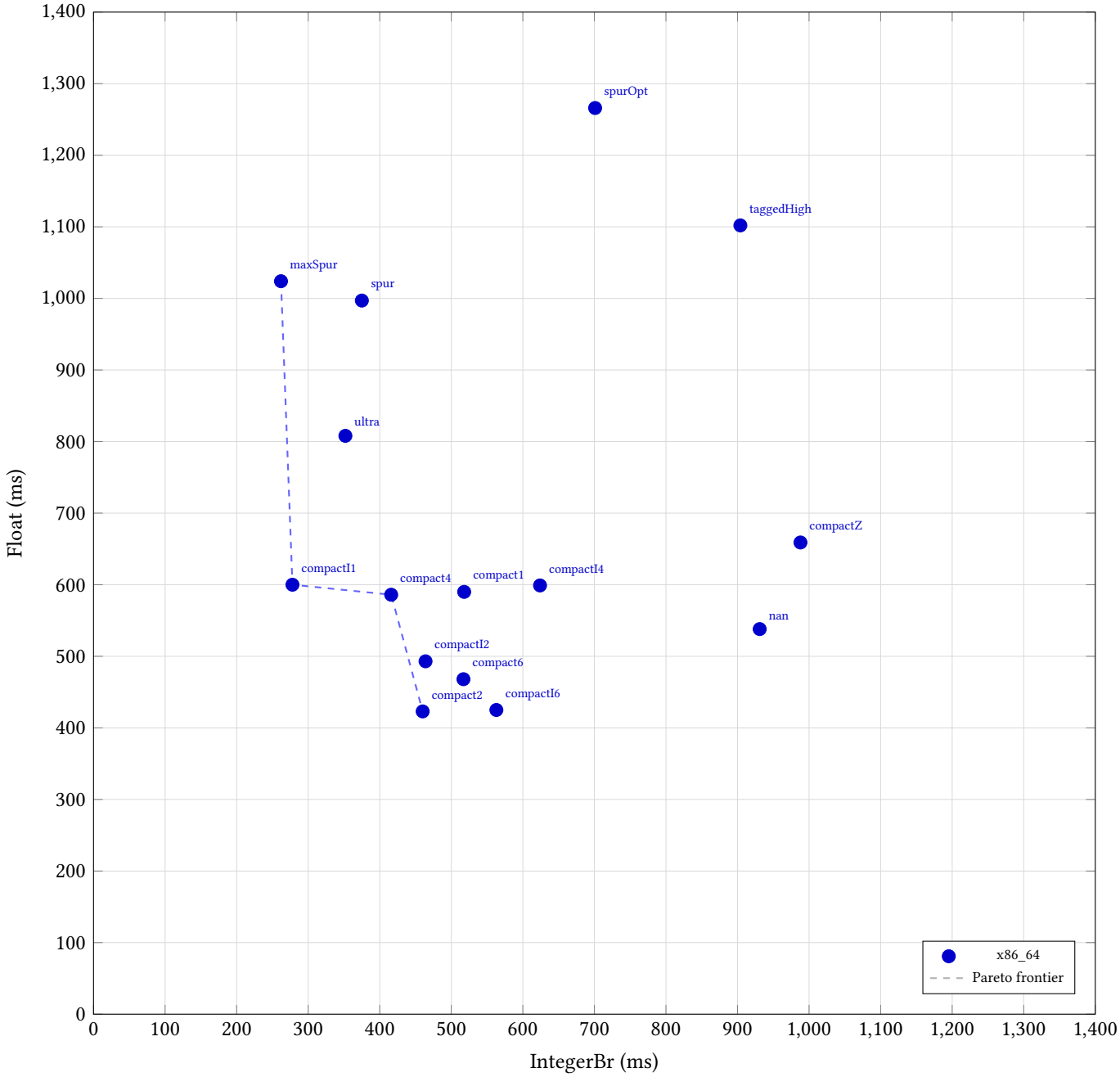

Figure 4. IntegerBr vs Float - X86-64

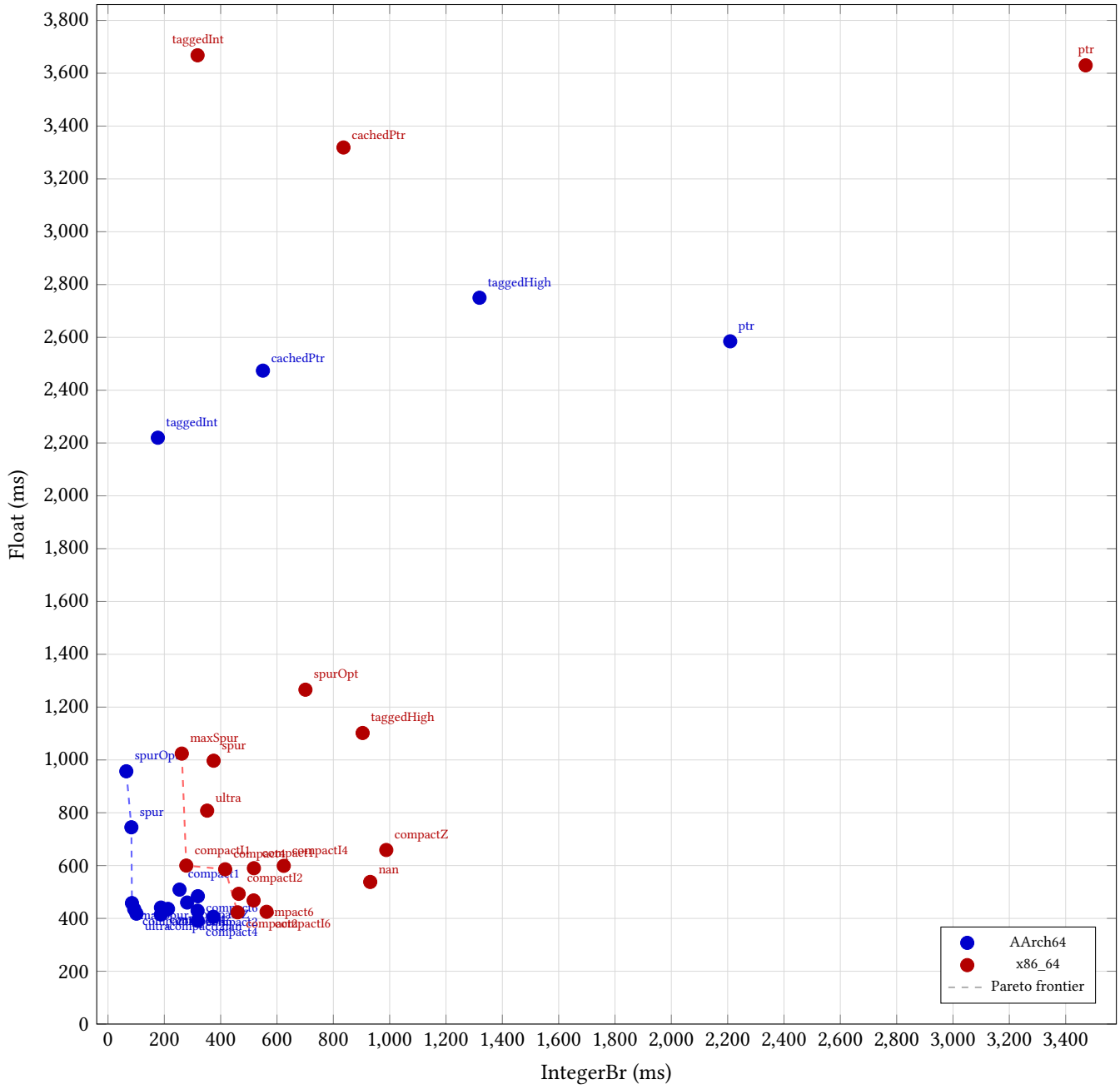

Figure 5. IntegerBr vs Float